

# What is an ethical engineer?

## Introduction

Engineering is not only about building structures, developing software, or solving technical issues; it is a profession that carries great responsibilities since engineers can shape the world and influence society, the environment, and future generations. That is why ethics plays such an important role in engineering; an ethical engineer is someone who demonstrates ethics, professional responsibility, and personal integrity through their decision-making. By going over the three main ethical theories—consequentialism, deontology, and virtue ethics—and highlighting the three qualities that characterize an ethical engineer—integrity, responsibility, and commitment—I will examine what it means to be an ethical engineer in this essay.

Engineering has an impact on almost every facet of human life in the modern world. The engineering decisions we make today will affect everything around us, from the streets and roads we walk on to the gadgets and devices we use every day. These choices often pose difficult ethical conundrums for engineers, such as balancing cost with public safety or quality and efficiency with environmental harm. The ever-increasing complexity of today's technological advances, such as artificial intelligence (AI) and global connectivity, further emphasizes the need for ethical engineering standards because these technological advances can have unforeseen negative impacts, and put people, communities, and ecosystems at risk if they are not supported by a solid ethical foundation.

## Ethical Theories and Engineering Practice

Ethical theories provide a solid foundation for understanding and evaluating ethical dilemmas in engineering. Three major ethical theories are consequentialism, deontology, and virtue ethics, and each of them offers a different perspective on what it means to act ethically.

### *Consequentialism*

According to the philosophy of consequentialism, the morality of an action is determined by the results it produces. This implies that ethical engineers need to think carefully about the implications of their choices and try to maximize the benefits while minimizing the disadvantages. For example, in order to guarantee the highest level of safety and dependability when designing a bridge, a consequentialist engineer would place a higher priority on choosing the best possible materials and the safest construction methods, even if these are more costly, more time-consuming, and more labor-intensive. The primary contention is that moral engineers have an obligation to create choices that are advantageous to all stakeholders. Another instance of consequentialist engineering thought is the development of renewable energy sources. Engineers working with solar and wind power need to weigh the environmental benefits of clean

energy against the costs of production and installation. A consequentialist strategy would place long-term viability and low environmental damage above short-term financial gain. However, consequentialism has its own drawbacks. It requires engineers to make accurate predictions, which can often be difficult due to unforeseen factors that can come into the play. Furthermore, unethical practices may be justified if they lead to positive overall outcomes. For example, a company may use less expensive, lower-quality materials in a particular product if it believes that the overall harm is less than the advantages of accessibility and cost savings. Ethical engineers should also be vigilant to prevent unethical shortcuts or reductions in public safety due to their consequential reasoning.

### *Deontology*

Deontology, which is developed by philosopher Immanuel Kant, states that ethics depend not only on consequences but also by doing assigned tasks, following personal obligations, and established rules. According to this point of view, an ethical engineer must do things by the industry standards, professional codes of conduct, and accepted ethical principles, even if doing so would result in dire consequences. For example, although skipping some testing may speed up the manufacturing process, an engineer working on a medical device must ensure that all safety and regulatory requirements are met. Deontological engineers value disciplined action under any circumstances. The aerospace industry's commitment to safety regulations is a practical illustration of deontological ethics in engineering standards. To ensure that aircraft and spacecraft meet strict safety standards, aerospace engineers must adhere to strict regulations set forth by organizations such as NASA and the FAA. If there is a cheaper, faster method of production, engineers must reject it if it violates their safety standards. However, ethics has its problems because it is not perfect. Strict adherence to rules can sometimes make it difficult for engineers to adapt to unexpected situations or come up with more creative solutions. When rules are outdated or do not take into account new technologies, engineers must carefully balance their ethical responsibilities with their drive to innovate and improve.

### *Virtue Ethics*

When comparing to strict rules and high standards or beneficial outcomes, virtue ethics emphasizes a person's moral principles and characteristics more. Ethical engineers who follow this idea strive to develop their personal traits such as honesty, courage, and wisdom. For example, an engineer working on environmental development projects demonstrates virtue stewardship by advocating for environmentally friendly solutions, even if it means going against the company's best interests. The virtues that engineers need to guide their moral behavior and decision-making can be developed through a process of continuous self-improvement and ethical reflection. Even in situations where regulations lack precise guidance, virtue ethics encourages engineers to develop moral character and behave in a disciplined manner. Even when there is no legal need, an honest engineer will disclose any potential hazards associated with a project.

Likewise, a fair-minded engineer will advocate for ethical labor standards and ensure that employees are treated fairly, even in the face of financial hardship.

## **Characteristics of an Ethical Engineer**

Ethical behavior in engineering can take many forms, but three key characteristics define an ethical engineer: integrity, responsibility, and commitment to public benefits. These qualities shape ethical decision-making and help building trust in the engineering profession.

### ***Integrity***

For any ethical engineer, integrity is an essential trait. It involves always prioritizing ethical issues before individual or business benefit and acting with transparency and honesty in every element of a profession. Integrity-driven engineers accurately disclose test results and prevent unethical behavior. Consider the Volkswagen emissions scandal, where engineers concealed the true pollution levels while manipulating emissions statistics to comply with standards. An ethical engineer, however, would have upheld integrity by refusing to falsify data, ensuring that public health and the environment came first. Integrity is important because trust is the cornerstone of engineering, and it is built on integrity. Clients, governments, and the public rely on engineers to create safe, functional, and ethical solutions. Without integrity, engineering decisions can be corrupted or negligent, leading to disasters such as structural failure or environmental damage.

### ***Responsibility***

In addition to their employers, ethical engineers should be clear about their obligations to society and the environment. In engineering, accountability requires taking responsibility for one's actions, proactively avoiding harm, and adhering to ethical principles. For example, when constructing a high-rise building, civil engineers must follow safety rules and use durable materials to avoid earthquake threats. If an engineer ignores these responsibilities and approves subpar materials in an attempt to save money, they put lives in danger. The collapse of the Rana Plaza garment factory in Bangladesh in 2013 is an example of what can happen when engineers and business leaders fail to uphold their responsibilities. Engineers should also be tasked with solving ethical puzzles within their companies. When public safety is at risk, it is sometimes necessary to speak up or even report to a whistleblower. To prevent harm, an ethical engineer who discovers dishonest and illegal business practices should take appropriate steps, whether that means filing an internal complaint or going to court.

### ***Commitment to Public Welfare***

The public interest must always take precedence over the interests of engineers, their organizations, or their political parties. This means going beyond simply choosing the easiest

path and actively seeking answers that minimize harm and improve society. For example, an ethical engineer designing an industrial facility would try to minimize waste by using environmentally friendly alternatives and increasing energy efficiency, even if these options would cost more time, money, and effort. Likewise, biomedical engineers who create medical devices, vaccines, and medicines need to ensure that their creations are not only safe, but also affordable and available to those who need them the most. There is also another emerging field where the public interest is placed on top: artificial intelligence (AI). Engineers who are involved in developing AI, also need to ensure that AI's algorithms not invade public privacy or have negative impacts on society.

## Why These Characteristics Matter

The importance of integrity, responsibility, and commitment to public welfare extends beyond individual engineers; these qualities shape the entire profession, benefiting society at large.

- **Trust and Credibility:** Integrity fosters public trust in engineering solutions, ensuring that technological advancements are reliable, safe, and beneficial.
- **Safety and Accountability:** Responsibility prevents disasters, holds engineers accountable, and promotes a culture of diligence and continuous improvement.
- **Sustainability and Social Justice:** Commitment to public welfare ensures that engineering contributes to long-term sustainability, equity, and societal well-being.

These characteristics also resonate with professional engineering codes of ethics, such as the National Society of Professional Engineers (NSPE) Code of Ethics, which emphasizes honesty, responsibility, and the public interest. Ethical engineers do not merely follow regulations; they actively shape the future of engineering by integrating moral considerations into their daily work.

## Conclusion

According to theories like consequentialism, deontology, and virtue ethics, an ethical engineer is the one who consistently maintains his or her personal traits such as integrity, responsibility, and dedication to the public benefits and interests. In engineering, these qualities are needed to maintain sustainability, safety, and mutual trust among others. The needs for ethical engineers nowadays is growing as engineering problems in general and specifically technology are becoming more complex and advanced. In addition to protecting the public, engineers should also maintain professional integrity and contribute to creating a better world for future generations by adhering to the ethical standards of the profession.